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Completion nipple identifier tool

Abstract

A tubing nipple impression device for use within downhole tubing includes a driver body, a stopping disk, first and second impression arms, and upper and lower radial springs. The driver body extends in an axial direction along a vertical axis of the downhole tubing. The driver body further includes a nose portion, and first and second driver arms that abut against the stopping disk. The shear pin attaches the driver body to the stopping disk. The first and second impression arms are fixed to the stopping disk, and are further connected to each other with upper and lower radial springs. The first and second impression arms are actuated by the nose portion to form first and second impressions complimentary to first and second portions of a tubing nipple, respectively, when the first and second impression arms abut against the tubing nipple.

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Background/Summary

BACKGROUND

(1) During the process of preparing a well for drilling operations, completion tubing is installed in the well to form a barrier between a subterranean formation and the wellbore. The completion tubing is typically formed of materials such as steel, and creates a conduit from the bottom of the wellbore, where production fluids may be found, to the surface of the Earth for further processing. A common part of well completion tubing is the nipple. A nipple is a completion component fabricated as a short section of thick tubular with a machined internal surface that provides a seal area and a locking profile for attaching various devices thereto. There are many types of nipples utilized in well completions based upon their required functionality, including No-Go Nipples, Selective-Landing Nipples, and Ported Safety Valve Nipples, for example.

(2) In well intervention operations and in order to install flow-control devices, such as plugs, chokes, or sub-service safety valves, the type of nipples must be identified prior to the job. In many cases, particularly for wells with outdated completion tubing, the type of nipple may not be accurately reflected in the drilling records. Furthermore, if a flow-control device with an incorrect

seating profile is installed, the flow-control device will become stuck, resulting in a loss of well production and potentially the entire well intervention string.

SUMMARY

(3) A tubing nipple impression device for use within downhole tubing includes a driver body, a stopping disk, first and second impression arms, and upper and lower radial springs. The driver body extends in an axial direction along a vertical axis of the downhole tubing. The driver body further includes a nose portion, and first and second driver arms that abut against the stopping disk. The shear pin attaches the driver body to the stopping disk. The first and second impression arms are fixed to the stopping disk, and are further connected to each other with upper and lower radial springs. The first and second impression arms are actuated by the nose portion to form first and second impressions complimentary to first and second portions of a tubing nipple, respectively, when the first and second impression arms abut against the tubing nipple.

(4) A method for creating an impression of a tubing nipple includes attaching a driver body to a stopping disk with a shear pin. The driver body extends in an axial direction along a vertical axis of downhole tubing and includes a nose portion, a first driver arm, and a second driver arm. The stopping disk abuts against the first driver arm and the second driver arm. The method further includes fixing a first impression arm and a second impression arm to the stopping disk. The first impression arm is connected to the second impression arm with an upper radial spring and a lower radial spring. The first impression arm and the second impression arm are then actuated towards the tubing nipple with the nose portion of the driver body, causing the first impression arm and the second impression arm to abut against the tubing nipple. In this way, the method further includes forming a first impression complimentary to a first portion of the tubing nipple, and forming a second impression complimentary to a second portion of the tubing nipple while the impression arms abut against the tubing nipple.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Specific embodiments of the disclosed technology will now be described in detail with reference to the accompanying figures. Like elements in the various figures are denoted by like reference numerals for consistency. The sizes and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not necessarily drawn to scale, and some of these elements may be arbitrarily enlarged and positioned to improve drawing legibility.

(2) FIG. 1 depicts a well site in accordance with one or more embodiments of the invention.

(3) FIG. 2 depicts a tubing nipple impression device in accordance with one or more embodiments of the invention.

(4) FIG. 3 depicts a tubing nipple impression device in accordance with one or more embodiments of the invention.

(5) FIG. 4 depicts a flowchart of a method in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(6) Specific embodiments of the disclosure will now be described in detail with reference to the accompanying figures. In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well known features have not been described in detail to avoid unnecessarily complicating the description.

(7) Throughout the application, ordinal numbers (e.g., first, second, third, etc.) may be used as an

adjective for an element (i.e., any noun in the application). The use of ordinal numbers is not intended to imply or create any particular ordering of the elements nor to limit any element to being only a single element unless expressly disclosed, such as using the terms “before”, “after”, “single”, and other such terminology. Rather, the use of ordinal numbers is to distinguish between the elements. By way of an example, a first element is distinct from a second element, and the first element may encompass more than one element and succeed (or precede) the second element in an ordering of elements.

(8) In addition, throughout the application, the terms “upper” and “lower” may be used to describe the position of an element in a wellbore as described herein. In this respect, the term “upper” denotes an element disposed closer to the surface of the Earth than a corresponding “lower” element when the tubing nipple impression device is in the wellbore, while the term “lower” conversely describes an element disposed further from the surface of the Earth than a corresponding “upper” element. Likewise, the term “axial” refers to an orientation substantially parallel to an extension direction of a completion tubular and wellbore, while the term “radial” denotes a direction orthogonal to an axial direction. Similarly, the terms “vertical” and “vertically” refer to an axial direction (i.e., the primary extension direction of the wellbore) while the terms “lateral” and “laterally” refer to the radial direction orthogonal to a vertical direction.

(9) In general, one or more embodiments of the disclosure are directed towards a device for creating an impression of an unknown nipple of completion tubing in a wellbore. The device includes a first impression arm and a second impression arm that abut against the nipple portion of the completion to create a partial impression of the nipple. More specifically, the first impression arm and the second impression arm each include a lead (or equivalent soft material) body that molds to the shape of the nipple. In this way, the first impression arm and the second impression arm replicate the unknown structure of the completion tubing nipple, which is analyzed at the surface by an engineer or operator to determine the type of nipple present.

(10) FIG. 1 shows a schematic diagram illustrating an example of a well site **11**. In general, well sites are configured in a myriad of ways. Therefore, the well site **11** is not intended to limit the particular configuration of the drilling equipment. For example, the well site **11** is depicted as being on land, however the well site **11** may be offshore and drilling may be carried out with or without the use of a marine riser. Moreover, various components and details of the well site that would be well known to a person of ordinary skill in the art have been omitted for the sake of brevity.

(11) A drilling operation at the well site **11** is initiated by drilling a wellbore **13**, or borehole, into a subterranean formation **15**. Once the wellbore **13** has been drilled, completion tubing **25** is cemented against the subterranean formation **15** to form a conduit for fluid transmission. The completion tubing **25** includes a tubing nipple **27**, which provides a sealing surface and locking profile for attaching downhole components thereto. Both the completion tubing **25** and the tubing nipple **27** may be formed of one or more varieties of steel (such as martensitic steel, duplex steel, or a steel alloy), and the tubing nipple **27** has a thickened wall to enhance its durability. Once the completion tubing **25** is installed, downhole operations may commence to acquire oil and gas from reserves in the wellbore **13**.

(12) However, and as noted above, a profile of a tubing nipple **27** may not be known. For example, if the schematics or logs of the wellbore **13** are incomplete, the tubing nipple **27** may not be properly marked. Alternatively, if the tubing nipple **27** becomes damaged from a well operation, the profile of the tubing nipple **27** will change. To address this, the slickline **19** is attached to a fishing neck (e.g., FIG. 3) of the tubing nipple impression device **23**. In this way, the tubing nipple impression device **23** is axially actuated throughout the wellbore **13** by the slickline **19** until the tubing nipple impression device **23** is able to capture a profile of the tubing nipple **27**. Once the tubing nipple impression device **23** has captured the profile of the tubing nipple **27**, the tubing nipple impression device **23** is lifted to the surface of the Earth by the slickline **19**. This allows a production engineer or operator to analyze the profile of the tubing nipple **27** captured by the

tubing nipple impression device **23**, which, in turn, allows the tubing nipple **27** itself to be identified. After the tubing nipple **27** is identified, well operations may resume by attaching downhole components to the tubing nipple **27** and/or documenting the geometry of the tubing nipple **27**.

(13) Turning to FIG. 2, FIG. 2 depicts a section view of a tubing nipple impression device **23** in accordance with one or more embodiments of the invention. The tubing nipple impression device **23** is formed of a plurality of components that serve to collectively actuate a first impression arm **55** and a second impression arm **61** in a lateral, or radial, direction. Structurally, the tubing nipple impression device **23** is primarily formed of a driver body **33**, a stopping disk **35**, and the first impression arm **55** and the second impression arm **61**. The driver body **33** and the stopping disk **35** may be formed, for example, of metals such as steel, iron, aluminum, an alloy, a combination thereof, or derivatives thereof, for example, or any other material suitable for downhole operations known to a person of ordinary skill in the art.

(14) The driver body **33** serves to actuate the first impression arm **55** and the second impression arm **61** of the tubing nipple impression device **23**. Specifically, the driver body **33** extends along a vertical axis **29** of the wellbore **13**, and includes a fishing neck **31**, a first driver arm **43**, a second driver arm **45**, and a nose portion **67**. The fishing neck **31** is a threaded connector that allows the driver body **33** to be fixed to a fishing tool (not shown) of a slickline **19**. The first driver arm **43** and the second driver arm **45** are angled protrusions that extend in a radial (or lateral) direction from the driver body **33**, and the nose portion **67** is a sloped surface that abuts against the first impression arm **55** and the second impression arm **61** to enlarge a radial distance therebetween as the driver body **33** is actuated. During operation, and as discussed below, the nose portion **67** is actuated in a downwards direction, and the abutment of the nose portion **67** to the first impression arm **55** and the second impression arm **61** causes the first impression arm **55** and the second impression arm **61** to expand in the wellbore **13** to contact the tubing nipple **27**.

(15) The first impression arm **55** and the second impression arm **61** are each formed with a head portion and a lead body that are rigidly fixed to each other. Specifically, the first impression arm **55** includes a first head portion **57** and a first lead body **59**, while the second impression arm **61** includes a second head portion **63** and a second lead body **65**. The first head portion **57** and the second head portion **63** are formed of a rigid metal, such as steel or iron, for example, while the first impression arm **55** and the second impression arm **61** are formed of a malleable metal such as lead or tin, for example.

(16) The first head portion **57** is fixed to the first lead body **59** and the second head portion **63** is fixed to the second lead body **65**, respectively, with a welding or brazing procedure, or using a mechanical fastener such as a bolt or screw, for example. This allows the first head portion **57** and the second head portion **63** to be actuated by the nose portion **67** of the driver body **33**, while the first lead body **59** and the second lead body **65** are actuated by the first head portion **57** and the second head portion **63**, respectively. That is, the driver body **33** abuts against the first head portion **57** and the second head portion **63** to actuate the first impression arm **55** and the second impression arm **61**, such that the first lead body **59** and the second lead body **65** do not need to be in contact with the driver body **33** to actuate. By actuating the first lead body **59** and the second lead body **65** with the first head portion **57** and the second head portion **63** rather than the driver body **33** itself, the first lead body **59** and the second lead body **65** retain their shape until they abut against the tubing nipple **27**. In this way, the first lead body **59** and the second lead body **65** remain undeformed from the motion of the driver body **33** while being actuated by the first head portion **57** and the second head portion **63**.

(17) To further aid in forming an impression of the tubing nipple **27**, the first lead body **59** and the second lead body **65** extend radially beyond the first head portion **57** and the second head portion **63**, respectively. This ensures that the first lead body **59** and the second lead body **65** contact the tubing nipple **27** without interference from the first head portion **57** or the second head portion **63**.

Similarly, the first head portion **57** and the second head portion **63** are fixed to the first lead body **59** and the second lead body **65** such that an air gap is present between the lead bodies and the driver body **33** while the driver body **33** abuts against the first head portion **57** and the second head portion **63**. This ensures that the driver body **33** is unable to deform the first lead body **59** or second lead body **65** during its actuation, as the driver body **33** is in contact with the first head portion **57** and the second head portion **63** instead. In addition, the second head portion **63** and the second lead body **65** are formed with a sloped internal profile that matches the profile of the first head portion **57** and the first lead body **59** to guide the motion of the driver body **33**. Such is depicted in FIGS. **2** and **3** by the sloped nature of the first lead body **59** to match the profile of the driver body **33**, and the sloped nature may also be applied to the second impression arm **61** as well. Alternatively, an internal profile of the first impression arm **55** and the second impression arm **61** may be sloped to match the driver body **33**, while an external profile of the first impression arm **55** and the second impression arm **61** is curved to match the stopping disk **35** and/or the completion tubing **25**.

(18) To maintain the abutment of the first head portion **57** and the second head portion **63** to the nose portion **67** of the driver body **33**, the tubing nipple impression device **23** further includes a plurality of radial springs that interconnect the first impression arm **55** to the second impression arm **61**. Specifically, an upper radial spring **51** connects the first head portion **57** to the second head portion **63**, while a lower radial spring **69** interconnects the first lead body **59** to the second lead body **65**. The upper radial spring **51** and the lower radial spring **69** deliver a compressive force to the first impression arm **55** and the second impression arm **61** that retains the arms to the driver body **33** while the driver body **33** is actuated downwards. In addition, the upper radial spring **51** and the lower radial spring **69** may surround the entirety of the tubing nipple impression device **23** such that the driver body **33** extends through the upper radial spring **51** and the lower radial spring **69**. Alternatively, the upper radial spring **51** and lower radial spring **69** may only extend partially around the circumference of the tubing nipple impression device **23**, or may be formed of multiple pieces to ease the assembly process of the tubing nipple impression device **23**.

(19) The first impression arm **55** and the second impression arm **61** are also indirectly connected to the driver body **33** prior to actuation. In particular, and as shown in FIG. **2**, the first head portion **57** is connected to a first pivot arm **47** with a first pivot pin **53**, and the second head portion **63** is connected to a second pivot arm **49** with a second pivot pin **71**. The first pivot pin **53** and the second pivot pin **71** may be embodied, for example, as a rod (or "pin") that rotates within a hole or orifice of the first head portion **57** and the second head portion **63**, respectively. The first pivot arm **47** and the second pivot arm **49** are formed of steel or an equivalent metal as described herein. At their other end, the first pivot arm **47** and the second pivot arm **49** are connected, with a third pivot pin **39** and a fourth pivot pin **41**, respectively, to a stopping disk **35** that is connected to the driver body **33** by way of a shear pin **37**. Thus, the first impression arm **55** and the second impression arm **61** are hinged to the stopping disk **35**, and can rotate away from the stopping disk **35** and the driver body **33** to abut against the completion tubing **25** and tubing nipple **27** depicted in FIG. **1**.

(20) Keeping with FIG. **2**, the stopping disk **35** is an annular block of metal that surrounds the upper portion of the driver body **33**, and serves to prevent the driver body **33** from actuating excessively vertically upwards during the process of creating an impression of the tubing nipple **27**. To this end, the driver body **33** includes a first driver arm **43** and a second driver arm **45** that extend radially therefrom, and the stopping disk **35** rests upon the first driver arm **43** and the second driver arm **45**. The stopping disk **35** is further connected to the driver body **33** by way of a shear pin **37** as noted above, which extends radially through the stopping disk **35** and the driver body **33**.

(21) To actuate the driver body **33**, the driver body **33** includes a fishing neck **31** formed as a threaded connection as described above. The fishing neck **31** is connected to a fishing tool (not shown) of a slickline **19** as depicted in FIG. **1**. During a downhole operation, the driver body **33** is actuated with a jarring motion of the slickline **19**, which shears the shear pin **37** and forces the driver body **33** away from the stopping disk **35**. The downward actuation of the driver body **33**

causes the sloped nose portion **67** to abut against the first impression arm **55** and the second impression arm **61**, which causes the first impression arm **55** and the second impression arm **61** to radially actuate.

(22) Turning to FIG. **3**, FIG. **3** depicts a section view of a tubing nipple impression device **23** in an actuated position according to one or more embodiments of the invention. As shown in FIG. **3**, the driver body **33** of the tubing nipple impression device **23** has been forced downward, breaking the shear pin **37** that interconnects the stopping disk **35** and the driver body **33**. As noted above, the downward force on the driver body **33** is created by a jarring or rapid movement by the slickline **19**, which is attached to the tubing nipple impression device **23** by way of a fishing tool (not shown) that is fixed to the fishing neck **31** of the driver body **33**. The downward force on the driver body **33** is transmitted through the sloped nose portion **67** to the first impression arm **55** and the second impression arm **61**. This force, in turn, causes the first impression arm **55** and the second impression arm **61** to rotate about the stopping disk **35** via the first pivot arm **47** and the second pivot arm **49**, respectively, until the first impression arm **55** and the second impression arm **61** contact the tubing nipple **27**.

(23) After the initial jarring force is applied to the driver body **33**, a continued force is applied to the driver body **33** to maintain contact between the first lead body **59** and the tubing nipple **27** and the second lead body **65** and the tubing nipple **27**. During this period of time, the first lead body **59** and the second lead body **65** form to the shape of a portion of the tubing nipple **27**, thereby creating a partial profile of the tubing nipple **27**. Because the tubing nipple **27** is a symmetrical and uniform cutout of the completion tubing **25**, the partial profiles captured by the first lead body **59** and the second lead body **65** are sufficient to identify the geometry of the tubing nipple **27** as a whole. Once the first lead body **59** and the second lead body **65** have conformed to the shape of the tubing nipple **27**, an upward force is applied to the driver body **33** via the slickline **19**, which causes the first impression arm **55** and the second impression arm **61** to withdraw from the tubing nipple **27**. That is, once the driver body **33** has an upward force applied thereto, the upper radial spring **51** and the lower radial spring **69** apply a compressive force to the first impression arm **55** and the second impression arm **61** causing their withdrawal from the tubing nipple **27**. With the first lead body **59** and the second lead body **65** dislodged from the tubing nipple **27**, the driver body **33** may be freely lifted through the completion tubing **25** to the surface, where the first lead body **59** and the second lead body **65** are analyzed to determine the profile of the tubing nipple **27**.

(24) FIG. **4** depicts a flowchart of a method for creating an impression of a tubing nipple consistent with one or more embodiments of the invention as described herein. The steps of FIG. **4** may be performed, for example, by the aforementioned tubing nipple impression device **23**, but are not limited thereto. Additionally, the constituent steps of the method depicted in FIG. **4** may be performed in any logical order, and are not limited to the sequence presented. Finally, steps of FIG. **4** may be performed simultaneously or as part a single action, or a single step may be performed with multiple actions.

(25) The method of FIG. **4** initiates with step **410**, in which a driver body **33** is attached to a stopping disk **35** with a shear pin **37**. The driver body **33** includes a nose portion **67** with an angled or otherwise sloped profile relative to a vertical axis **29**. In addition, the driver body **33** includes an integrally formed first driver arm **43** and second driver arm **45** that laterally extend away from the driver body **33** towards the completion tubing **25**. The first driver arm **43** and the second driver arm **45** provide a shelf for the stopping disk **35** to rest upon, which eases the stress placed on the shear pin **37** while the tubing nipple impression device **23** is lowered through the completion tubing **25**. The shear pin **37** is attached to the driver body **33** and the stopping disk **35** by inserting the shear pin **37** into orifices that extend through the driver body **33** and the stopping disk **35**. Specifically, each of the driver body **33** and the stopping disk **35** are formed with an orifice having the same diameter as the shear pin **37**, and the orifices are aligned while the stopping disk **35** rests on the driver body **33** such that the shear pin **37** may be inserted therethrough. Once the shear pin **37** has

been inserted through the driver body **33** and the stopping disk **35**, the method proceeds to step **420**.

(26) In step **420**, a first impression arm **55** and a second impression arm **61** are interconnected by an upper radial spring **51** and a lower radial spring **69**, and are attached to the stopping disk **35**. It is noted that step **410** and step **420** may be performed in any order, or as part of a single step, as steps **410** and **420** are directed towards the assembly of a tubing nipple impression device **23** overall. During step **420**, the first impression arm **55** and the second impression arm **61** are connected by brazing or welding (or equivalent procedure) an upper radial spring **51** to a first head portion **57** and a second head portion **63** of the first impression arm **55** and the second impression arm **61**, respectively. Similarly, a lower radial spring **69** is fixed to the lowermost portions of the first lead body **59** and the second lead body **65**. The upper radial spring **51** and the lower radial spring **69** collectively serve to withdraw the first impression arm **55** and the second impression arm **61** from the tubing nipple **27** after an impression is formed with the first lead body **59** and the second lead body **65**, as described above.

(27) The first impression arm **55** and the second impression arm **61** are attached to the stopping disk **35** by way of a first pivot arm **47** and a second pivot arm **49**. Specifically, the first pivot arm **47** is attached to the first impression arm **55** by way of a first pivot pin **53**, and is further attached to the stopping disk **35** by way of a third pivot pin **39**. Similarly, the second pivot arm **49** is connected to the second impression arm **61** by way of a second pivot pin **71**, and is further connected to the stopping disk **35** by way of a fourth pivot pin **41**. Each of the first pivot pin **53**, the second pivot pin **71**, the third pivot pin **39** and the fourth pivot pin **41** may be embodied as a pin and hole joint that allow the first impression arm **55** and the second impression arm **61** to be connected to the stopping disk **35** in a hinged fashion. In turn, this allows the first impression arm **55** and the second impression arm **61** to maintain a fixed distance from the stopping disk **35**, and ensures that a uniform impression of the tubing nipple **27** is created. Once the first impression arm **55** and the second impression arm **61** are fixed to the stopping disk **35**, the method proceeds to step **430**.

(28) In step **430**, the first impression arm **55** and the second impression arm **61** are actuated towards the tubing nipple **27**. In this step, the driver body **33** is forced downwards by a quick, jarring motion of the slickline **19**. This causes the shear pin **37** to shear and release the driver body **33** from the stopping disk **35**. The driver body **33** is wedged between the first impression arm **55** and the second impression arm **61**, which causes the first impression arm **55** and the second impression arm **61** to actuate in a radial direction towards the completion tubing **25** and the tubing nipple **27**. At this point, the method proceeds to step **440**.

(29) In step **440**, the first impression arm **55** is abutted against the tubing nipple **27** to form the first lead body **59** into a first shape complimentary to a first portion of the tubing nipple **27**. Specifically, during this step the driver body **33** continues to press on the first head portion **57** of the first impression arm **55**, and the first lead body **59** that is fixed to the first head portion **57** is pressed into the tubing nipple **27** as a consequence. As noted above, the first lead body **59** is attached to the first head portion **57** with a brazing or welding process, or an adhering mechanism such as rivets, screws, or bolts. In addition, the first lead body **59** is positioned such that the first lead body **59** extends further towards the tubing nipple **27** than the first head portion **57** in an axial direction, which allows the first lead body **59** to abut against the tubing nipple **27** prior to the first head portion **57** abutting against the completion tubing **25**. Due to the first lead body **59** being formed of a malleable material such as lead, tin, or silver, for example, the first lead body **59** molds to the shape of the tubing nipple **27** while abutted. This creates an impression of a portion of the tubing nipple **27**, which is representative of the entirety of the tubing nipple **27** due to the symmetrical nature thereof.

(30) In step **450**, the second impression arm **61** is also abutted against the tubing nipple **27** to create a second shape complimentary to a second portion of the tubing nipple **27**. It is noted that steps **440** and **450** may be performed by a single actuation of the driver body **33**, and may occur

simultaneously as a result. During step 450, and similar to step 440, the driver body 33 is actuated downwards and abuts against the second head portion 63 of the second impression arm 61, which causes the second impression arm 61 to move in a radial direction to abut the second lead body 65 against the tubing nipple 27. As the second lead body 65 is pressed into the tubing nipple 27, the second lead body 65 molds to the shape of the tubing nipple 27 to form a second impression complimentary to a second portion of the tubing nipple 27. The first impression and the second impression may be compared to determine if the tubing nipple 27 is damaged or uniform, and the redundancy of creating a second impression avoids the need for a second downhole trip if the first impression is not properly formed or becomes damaged. Once the first and the second impression have been created by the first lead body 59 and the second lead body 65, the driver body 33 is lifted by the slickline 19, which causes the first impression arm 55 and the second impression arm 61 to withdraw towards the vertical axis 29 by way of the upper radial spring 51 and the lower radial spring 69. The tubing nipple impression device 23 is then lifted to the surface of the Earth for further analysis of the first impression arm 55 and the second impression arm 61, which completes the tubing nipple 27 impression creation process.

(31) Accordingly, the aforementioned embodiments of the invention as disclosed relate to devices and methods useful to create an impression of a tubing nipple of a wellbore. In addition, embodiments of the invention are capable of simultaneously creating multiple impressions of different portions of a tubing nipple, which allows the uniformity of the tubing nipple as a whole to be determined by comparing differences in the impressions. Furthermore, embodiments of the invention receive the benefits of a compact design, as the upper and lower radial springs ensure that the impression arms remain withdrawn while the tubing nipple impression device is raised and lowered within the wellbore.

(32) Although only a few embodiments of the invention have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. For example, the first and the second impression arms may be operated with a motor rather than requiring the driver body to be actuated. In addition, the driver body may be formed with a curved or rounded nose portion to change the actuation path of the impression arms. Furthermore, the internal and external profile of the first impression arm and the second impression arm may vary as described herein, and the first impression arm may be symmetrical to the second impression arm or vary in design. Finally, the tubing nipple impression device may be lowered into the wellbore with a drill string rather than a slickline. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims.

Claims

1. A tubing nipple impression device for use within downhole tubing, the device comprising: a driver body that extends in an axial direction along a vertical axis of the downhole tubing, the driver body comprising a nose portion, a first driver arm, and a second driver arm; a stopping disk that abuts against the first driver arm and the second driver arm of the driver body; a shear pin that attaches the driver body to the stopping disk; a first impression arm fixed to the stopping disk that is configured to form a first impression complimentary to a first portion of a tubing nipple when the first impression arm abuts against the tubing nipple; a second impression arm fixed to the stopping disk that is configured to form a second impression complimentary to a second portion of the tubing nipple when the second impression arm abuts against the tubing nipple; an upper radial spring that connects the first impression arm to the second impression arm; and a lower radial spring that connects the first impression arm to the second impression arm, wherein the nose portion of the driver body is configured to actuate the first impression arm and the second impression arm towards the tubing nipple to create an impression of the tubing nipple.

2. The tubing nipple impression device of claim 1, wherein the nose portion of the driver body has a sloped profile such that the nose portion is configured to enlarge a radial distance between the first impression arm and the second impression arm as the driver body is actuated.
3. The tubing nipple impression device of claim 1, wherein the upper radial spring and the lower radial spring are configured to actuate the first impression arm and the second impression arm after the impression is created to dislodge the first impression arm and the second impression arm from the tubing nipple.
4. The tubing nipple impression device of claim 1, wherein the first impression arm is fixed to the stopping disk with a first pivot arm and the second impression arm is fixed to the stopping disk with a second pivot arm.
5. The tubing nipple impression device of claim 4, wherein the first impression arm comprises a first head portion configured to be fixed to the first pivot arm, and a first lead body that is malleable to form the impression complimentary to the first portion of the tubing nipple when the first head portion abuts against the tubing nipple.
6. The tubing nipple impression device of claim 4, wherein the second impression arm comprises a second head portion configured to be fixed to the second pivot arm, and a second lead body that is malleable to form the impression complimentary to the second portion of the tubing nipple when the second head portion abuts against the tubing nipple.
7. The tubing nipple impression device of claim 1, further comprising: a threaded connector configured to connect the driver body to a slickline, the slickline being configured to actuate the driver body in the axial direction.
8. The tubing nipple impression device of claim 1, wherein the first driver arm and the second driver arm extend in a radial direction from the driver body.
9. The tubing nipple impression device of claim 1, wherein the upper radial spring is configured to be attached to a first head portion of the first impression arm and a second head portion of the second impression arm.
10. The tubing nipple impression device of claim 1, wherein the lower radial spring is configured to be attached to a first lead body of the first impression arm and a second lead body of the second impression arm.
11. A method for creating an impression of a tubing nipple, the method comprising: attaching, with a shear pin, a driver body to a stopping disk, wherein the driver body extends in an axial direction along a vertical axis of downhole tubing and comprises a nose portion, a first driver arm, and a second driver arm, and wherein the stopping disk abuts against the first driver arm and the second driver arm; fixing a first impression arm and a second impression arm to the stopping disk, and connecting the first impression arm to the second impression arm with an upper radial spring and a lower radial spring; actuating the first impression arm and the second impression arm towards the tubing nipple with the nose portion of the driver body; abutting the first impression arm against the tubing nipple to form a first impression complimentary to a first portion of the tubing nipple; and abutting the second impression arm against the tubing nipple to form a second impression complimentary to a second portion of the tubing nipple.
12. The method of claim 11, wherein actuating the first impression arm and the second impression arm to create the impression comprises: abutting a sloped profile of the nose portion against the first impression arm and the second impression arm and actuating the driver body, thereby enlarging a radial distance between the first impression arm and the second impression arm as the driver body is actuated.
13. The method of claim 11, further comprising: actuating the first impression arm and the second impression arm with the upper radial spring and the lower radial spring after the impression is created and dislodging the first impression arm and the second impression arm from the tubing nipple.
14. The method of claim 11, further comprising: fixing the first impression arm to the stopping disk

with a first pivot arm and fixing the second impression arm to the stopping disk with a second pivot arm.

15. The method of claim 11, further comprising: forming the impression complimentary to the first portion of the tubing nipple by abutting a malleable first lead body of the first impression arm against the tubing nipple.

16. The method of claim 11, further comprising: forming the impression complimentary to the second portion of the tubing nipple by abutting a malleable second lead body of the second impression arm against the tubing nipple.

17. The method of claim 11, further comprising: connecting the driver body to a slickline with a threaded connector and actuating the driver body in the axial direction with the slickline.

18. The method of claim 11, wherein the first driver arm and the second driver arm extend in a radial direction from the driver body.

19. The method of claim 11, further comprising: attaching the upper radial spring to a first head portion of the first impression arm and a second head portion of the second impression arm.

20. The method of claim 11, further comprising: attaching the lower radial spring to a first lead body of the first impression arm and a second lead body of the second impression arm.
